Esophageal Function Tests in Children: pH and Impedance Monitoring, High-Resolution Manometry and EndoFLIP.

Imagine three referrals arriving in the same week. The first is a three-month-old who spits up after most feeds but is growing well, and whose parents bring a report from another hospital that calls a pH study abnormal. The second is a twelve-year-old with heartburn that has not gone away on a proton pump inhibitor, the acid-suppressing medicine usually shortened to PPI, and whose upper endoscopy and biopsies were normal. The third is a six-year-old with difficulty swallowing whose esophageal manometry report reads diffuse esophageal spasm. Each family wants to know what the test means and what comes next.

The clinical question of this chapter is how to choose, order and interpret the three main esophageal function tests in children: combined multichannel intraluminal impedance and pH monitoring, usually shortened to pH-impedance or MII-pH; high-resolution manometry, shortened to HRM; and the functional lumen imaging probe, known by its commercial name EndoFLIP or simply FLIP. The main idea is this. None of these tests has validated pediatric normal values, and each answers a narrow question well. pH-impedance tells you whether reflux is present, how high it travels, whether it is acid or not, and whether symptoms follow it. Manometry tells you whether the body of the esophagus contracts and whether the junction with the stomach relaxes. FLIP tells you how stiff the esophageal wall is and how well the junction opens when it is stretched. Used loosely, the same tests turn normal infant reflux into a disease and turn normal small children into patients with spasm or obstruction.

We will move through the topic in order. First, how impedance works and what it adds to pH. Then what healthy infants actually look like on these studies, why there are no true pediatric normal values, and which working thresholds you can use. Then symptom association, the question of whether the study predicts the result of anti-reflux surgery, and how to do the study well. Then manometry under the Chicago Classification version four, why adult criteria over-call disease in children, and what the pediatric achalasia data show. Then FLIP, its evidence in adults and in pediatric eosinophilic esophagitis, and when it is worth doing. We finish with how to choose the test, what the guidelines say, the traps, and what to do differently in clinic.

A note on sources. For reflux testing we use the 2018 joint pediatric gastroesophageal reflux guideline written by Rosen and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. We also use the 2012 ESPGHAN standard protocol from the European Pediatric Impedance Group, known as EURO-PIG and written by Wenzl and colleagues, the 2017 position

statement of the British Society of Paediatric Gastroenterology, Hepatology and Nutrition, and, for adult thresholds, the Lyon Consensus version two published by Gyawali and colleagues in 2024. For manometry we use the Chicago Classification version four, published by Yadlapati and colleagues in 2021, and the pediatric studies by Singendonk and colleagues. For FLIP we use the clinical practice updates of the American Gastroenterological Association, known as the AGA, and the adult and pediatric studies named as we go.

Part one. How impedance works.

A pH probe measures one thing: the acidity at its tip. Impedance measures something different. Along the catheter sit pairs of metal rings, and between each pair a tiny alternating current flows. The resistance to that current is the impedance. Air conducts poorly, so it raises impedance. Liquid conducts well, so when a liquid bolus passes a pair of rings, impedance falls. By convention a reflux event is a fall of more than fifty percent from the baseline value. Because there are several pairs of rings along the catheter, the direction of the fall tells you which way the bolus moved. A fall that starts low and travels upward is reflux. A fall that starts high and travels downward is a swallow.

The pH sensor then classifies each event. If the pH falls below four, the event is acid. If it stays between four and seven, it is weakly acidic. If it is above seven, it is weakly alkaline. Finally, the resting impedance of the esophageal lining itself carries information. When the mucosa is damaged by reflux and becomes leaky, the baseline impedance during the night falls. This measure is called the mean nocturnal baseline impedance.

Why does this matter in children? According to the 2018 guideline, non-acid reflux makes up between forty-five and eighty-nine percent of pediatric reflux episodes, and a pH probe cannot see any of it. Infants are fed every few hours, and milk buffers the stomach for much of the day, so much of their reflux is not acid at all. Impedance adds the number of events, how high they go, how quickly the esophagus clears them, and a signal of mucosal integrity. It does not measure volume, and it is read partly by eye, so it is not entirely objective.

Part two. What the healthy infant looks like, and why there are no normal values.

The most striking data come from healthy preterm babies. In 2006, Lopez-Alonso and colleagues studied twenty-one healthy, asymptomatic preterm infants at a median of twelve days of life. They had a median of seventy-one reflux events in twenty-four hours. About seventy-three percent of events were weakly acidic, about twenty-five percent were acid, and none were weakly alkaline. Ninety percent of events reached the upper esophagus. The median acid exposure was about five and a half percent of the time, with a range from almost zero to nearly twenty-one percent. In other words, one healthy infant spent more than a fifth of the day with acid in the esophagus and had no symptoms. The implication is simple. An abnormal event count or acid time in an infant may be physiology, and a study cannot diagnose reflux

disease in an infant on numbers alone.

The 2018 guideline states this directly: no true pediatric normative values have been established, because it is not ethical to place catheters in healthy children for research. The best reference data come from children who were tested for symptoms and whose studies were judged negative. In 2020, Cresi and colleagues reported such a cohort of one hundred and ninety-five children. The ninety-fifth percentile of the reflux index, which is the percentage of time with pH below four, was about five point nine percent in newborns, four point seven percent in infants and two point five percent in children aged one to sixteen years. For impedance events, the ninety-fifth percentile was about five point seven events per hour in newborns, four point nine in infants and four point four in older children. The authors themselves describe their cohort as not exactly normal. For comparison, the adult Lyon Consensus calls an acid exposure time below four percent normal and above six percent conclusive for reflux disease, but those numbers were derived in adults and never validated in children.

So which numbers can you use? For acid exposure, the 2018 guideline quotes a pH below four for more than ten percent of the time as abnormal in infants under one year and more than five percent above one year. The German Pediatric Impedance Group, in a study published by Pilic and colleagues in 2011, used working thresholds of more than one hundred reflux events in twenty-four hours under one year and more than seventy events from one year of age. The adult Lyon thresholds are fewer than forty events arguing against reflux disease and more than eighty supporting it. For mean nocturnal baseline impedance, adults below one thousand five hundred ohms support reflux disease and above two thousand five hundred ohms argue against it, while in children a systematic review found only one study, with a cut-off of about one thousand four hundred and sixty-six ohms and modest accuracy. A newer measure called the post-reflux swallow-induced peristaltic wave index has been retired to research use by Lyon version two and has no pediatric cut-off. The practical rule is to write the age-specific threshold you used in the report, because abnormal by adult criteria is not the same statement as abnormal for this child.

Part three. What impedance adds, and whether symptoms follow reflux.

The German multicenter study by Pilic and colleagues gives the clearest picture of what impedance adds. Among seven hundred children aged three weeks to sixteen years, two hundred and seventy studies were abnormal. Forty-five percent were abnormal by impedance only, thirty-seven percent by both impedance and pH, and only eighteen percent by pH only. Infants had more reflux episodes than older children, as you would expect from their feeding volume and posture. The caveat is that abnormal by impedance only rests on event-count thresholds that are themselves not validated in children.

The second job of the study is to link symptoms to reflux. Two indices are used. The symptom index, described by Wiener and colleagues in 1988, is the percentage of symptoms that are preceded by reflux within a short window, and it is called positive at fifty percent or more. Its weakness is that it ignores how many reflux events occurred without any symptom. The symptom association probability, described by Weusten and colleagues in 1994, divides the recording into two-minute segments and applies a Fisher exact test to ask whether symptoms and reflux occur together more often than chance. It is expressed as one minus the P value, times one hundred, and is called positive at ninety-five percent or more. In the original series of one hundred and eighty-four studies, the two indices disagreed in about eleven percent. In the German pediatric study they agreed in eighty-two percent of studies, which means they disagreed in about one study in five.

Children add a further problem. The 2018 guideline notes that parents and patients fail to report more than half of symptoms, that cough and arching are not specific, and that no index has been shown to predict outcome, so it recommends no index over another. Even in adults, performance is modest: in one analysis the symptom index had a sensitivity of eighty percent and a specificity of only thirty-five percent, and the association probability a sensitivity of seventy-three percent and specificity of sixty-five percent. Lyon version two treats both as supportive only. So use symptom association in a verbal child with a frequent, well-marked symptom, or in an infant whose cough or apnea can be logged by a nurse or on video. Do not use a positive index alone to justify a fundoplication, and do not dismiss reflux because a sleepy infant had too few marked symptoms.

Part four. Does the study predict the operation?

Two pediatric surgical cohorts give conflicting answers. In 2010, Rosen and colleagues studied thirty-four children who underwent fundoplication. No preoperative pH-impedance marker, whether acid events, non-acid events, total events or reflux time, predicted the symptom outcome, and a positive symptom index or symptom sensitivity index did not predict improvement either. In 2017, Mauritz and colleagues followed twenty-five children after laparoscopic fundoplication. Acid exposure fell from eight point five percent to under one percent, the lower esophageal sphincter pressure rose from about ten to twenty-four millimeters of mercury, and children with more reflux events before surgery had a bigger fall afterwards. Three of the twenty-five developed new difficulty swallowing.

How do we read the two together? Fundoplication reliably lowers measured reflux. Whether the symptom that prompted surgery goes away is a separate question, and the preoperative study does not answer it. In practice, use pH-impedance to document that reflux is real and frequent, then make the surgical decision on the clinical picture, failed medical therapy and the family's goals.

Part five. Doing the study well.

The 2018 guideline suggests pH-impedance for two purposes: to correlate persistent troublesome symptoms with acid and non-acid reflux, and, when endoscopy is normal, to separate non-erosive reflux disease, shortened to NERD, from reflux hypersensitivity and functional heartburn. It can also clarify the role of reflux in esophagitis and test whether acid suppression is working. The EURO-PIG protocol adds that the test adds most for symptoms on PPI, for feeding-related symptoms and for intermittent symptoms, and that its limits are cost, a limited effect on treatment decisions, and the absence of evidence-based pediatric parameters.

The single most important decision is PPI status, and it should be made before the catheter is placed. Test off acid suppression when you want to know whether reflux causes the symptoms, and on it when you want to know whether therapy is working. A washout of about seven days off the PPI is common practice, but no pediatric guideline specifies a number. Record for about twenty-four hours, and for at least eighteen. Log meals, position and symptoms. Wireless pH capsules in older children record for at least forty-eight hours, and the 2018 guideline reports complication rates from zero to fifteen percent. In the report, state the age, the PPI status, the thresholds used and the symptom indices with their caveats. And do not test a thriving infant with regurgitation, or try to diagnose reflux disease in an infant from numbers alone.

Part six. High-resolution manometry and the Chicago Classification.

High-resolution manometry uses a catheter with closely spaced pressure sensors from the throat to the stomach, displaying pressure as a color map in space and time. The Chicago Classification version four, published in 2021, standardized both the protocol and the metrics, all derived in adults. The protocol includes ten supine swallows of five milliliters of water plus multiple rapid swallows, then at least five upright swallows plus a rapid drink challenge of two hundred milliliters. The provocative and positional tests reduce false calls.

Four metrics carry most of the diagnoses. The integrated relaxation pressure, or IRP, tells you whether the junction relaxes with a swallow. Its upper limit in the supine position is fifteen millimeters of mercury on Medtronic systems and twenty-two on Laborie or Diversatek systems, and in the upright position twelve or fifteen respectively, so the limit depends on the device. The distal contractile integral, or DCI, measures the vigor of the contraction: below one hundred is a failed swallow, from one hundred to four hundred and fifty is weak, and above eight thousand is hypercontractile. The distal latency measures timing: a latency under four and a half seconds with a DCI of at least four hundred and fifty is premature, a sign of lost inhibition as in spasm or type three achalasia. Panesophageal pressurization in at least twenty percent of swallows defines type two achalasia.

The diagnoses follow. Type one achalasia has a high IRP and one hundred percent failed peristalsis. Type two has a high IRP, one hundred percent failed peristalsis and panesophageal pressurization in at least twenty percent of swallows; it has the best response to treatment.

Type three has a high IRP, premature contractions in at least twenty percent of swallows, and no normal peristalsis. Esophagogastric junction outflow obstruction requires a high IRP both supine and upright with peristalsis still present, and, new in version four, it is conclusive only with relevant symptoms such as dysphagia or chest pain plus at least one supportive test, a timed barium esophagram or FLIP. That change was made because a raised IRP alone was over-called. Ineffective esophageal motility is defined as more than seventy percent ineffective swallows or at least fifty percent failed swallows, with fifty to seventy percent ineffective called inconclusive. Distal esophageal spasm and hypercontractile esophagus now also require symptoms. Note what manometry is not for: the 2018 pediatric guideline advises against using it to diagnose reflux disease, and in favor of it when a motility disorder is suspected.

Part seven. Children are not small adults.

The esophagus of a small child is shorter and narrower, and the metrics shift with size. In 2014, Singendonk and colleagues studied seventy-six children with a mean age of nine. Younger and smaller children had a higher IRP and a shorter distal latency, which are exactly the two metrics that define outflow obstruction and spasm. With standard adult criteria, sixty-six percent of the children were labeled as having abnormal motility. After age adjustment the figure fell to fifty percent, and after size adjustment to fifty-three percent. Outflow obstruction fell from thirteen percent to seven percent, and diffuse esophageal spasm fell from fourteen percent to one percent. That study used the earlier 2012 version of the Chicago criteria, and we found no pediatric consensus or validated pediatric normal values for version four. So treat spasm and outflow obstruction in a small child as hypotheses to be tested, not diagnoses.

Pediatric achalasia is rare but rising. In the United Kingdom, Marlais and colleagues found an incidence of zero point one eight cases per one hundred thousand children per year, from two hundred and twenty-eight cases between 1998 and 2008, rising by about twelve percent per year in odds. The mean age at diagnosis was about eleven years, and fifty-six percent were boys. Reading the tracings is harder than it looks. In a 2017 study by Singendonk and colleagues, six raters reviewed forty pediatric recordings. Agreement on the diagnosis of achalasia, measured by the kappa statistic, was zero point five two using the software alone and rose to zero point nine two with expert review. Agreement on subtype rose only from zero point two eight to zero point five eight. The authors recommend interpretation in expert pediatric motility centers. Finally, think beyond achalasia. In a 2020 series of fifty-four children with dysphagia studied by manometry, forty-eight percent had a motility disorder, and ineffective motility was as common as achalasia. Eosinophilic esophagitis and repaired esophageal atresia are the usual backdrops.

Part eight. EndoFLIP: measuring how the junction opens.

FLIP is a catheter carrying a sixteen-centimeter balloon that is placed across the esophagogastric junction during endoscopy and filled in steps, typically between thirty and sixty milliliters. Impedance electrodes inside the balloon compute the diameter along its length, and a sensor records the pressure. Dividing the narrowest cross-sectional area by the balloon pressure gives the distensibility index, shortened to DI, in square millimeters per millimeter of mercury. Watching the balloon over time, a technique called panometry, shows whether the body of the esophagus contracts in response to distension.

The adult evidence is strong. In 2016, Carlson and colleagues found that FLIP was abnormal in ninety-five percent of adults with abnormal manometry, including all seventy with achalasia, but also abnormal in half of those with normal manometry. In 2021 the same group studied five hundred and thirty-nine adults and found that a normal FLIP went with normal manometry or ineffective motility in ninety-five percent, while reduced junction opening with an absent or impaired contractile response matched an outflow disorder in ninety-two percent. In the 2025 AGA update, normal junction opening is a distensibility index of two point zero or more with a maximum diameter of sixteen millimeters or more, and reduced opening is a distensibility index below two point zero with a diameter below twelve millimeters. After treatment aimed at the sphincter, a distensibility index below two to three suggests incomplete treatment.

In children, the best data come from eosinophilic esophagitis. In 2017, Menard-Katcher and colleagues compared eighty-eight children with eosinophilic esophagitis with forty-four controls. Distensibility rose with age in controls and was lower in eosinophilic esophagitis, more so in children with fibrosis, strictures, rings and food impaction. In 2022, Hoffmann and colleagues studied fifty-nine patients aged nine to twenty-one. The median distensibility index was three point three in those with fibrotic features and five point five in those with inflammatory features. A distensibility index below four and a half predicted moderate rings with an area under the curve of zero point eight one, distensibility predicted food impaction after adjustment, and distensibility did not correlate with the eosinophil count. That last finding matters: stiffness and inflammation are separate axes of the disease.

So when is FLIP needed? There are four good questions. First, to confirm or refute outflow obstruction suggested by manometry, as the Chicago version four supportive test, with a timed barium swallow as the alternative. Second, during and after treatment of achalasia, to see whether the junction now opens. Third, in eosinophilic esophagitis, to judge remodeling and stricture risk and to follow therapy, though never to diagnose the disease, which requires biopsies. Fourth, after surgery, such as fundoplication or repair of esophageal atresia, to separate a tight wrap or narrow junction from dysmotility; in one small atresia series the median junction distensibility was about two point six. The limits are real. There are no pediatric FLIP normal values, the effect of sedation and anesthesia has not been studied, and the AGA advises never to base a diagnosis or treatment decision on FLIP alone and not to use it routinely for reflux.

Part nine. Choosing the test, and what the guidelines say.

Match the question to the tool. If the question is whether symptoms on PPI are still caused by reflux, the first test is pH-impedance on PPI, with endoscopy if it has not been done. If the question is whether a nonverbal child's cough or apnea is linked to reflux, do pH-impedance off PPI with nurse or video marking, and treat the association rather than the event count. For dysphagia with a normal endoscopy and biopsies, start with manometry, and add a timed barium swallow or FLIP if the IRP is high. If manometry shows outflow obstruction, confirm with FLIP or barium before treating, and reinterpret against pediatric data. For eosinophilic esophagitis with impaction or narrowing, consider endoscopy with FLIP, dilation if needed, and tracking distensibility on therapy. Before fundoplication, use pH-impedance to confirm reflux, and manometry if there is dysphagia or a suspected motility disorder.

On the guidelines, North America and Europe are largely aligned on reflux testing, because the main pediatric guideline is a joint one. Both the 2018 joint guideline and the European documents describe pH-impedance as the test for linking symptoms to reflux, both state that no true pediatric normal values exist, and the 2018 guideline prefers no symptom index while the adult Lyon consensus calls them supportive only. They differ on acid exposure thresholds: the pediatric guideline quotes ten percent for infants and five percent for older children, while the adult Lyon thresholds of four and six percent are sometimes applied to adolescents without validation. The material gap is manometry. There are no pediatric criteria, so the adult Chicago version four thresholds are applied to children, and the pediatric data show that this can manufacture spasm or outflow obstruction in a small child. For FLIP, the AGA treats it as an adjunct in adults, and there is no ESPGHAN statement and no pediatric normal range.

Part ten. Traps and pearls.

Four traps stand out. The first is the abnormal pH study in an infant: healthy infants reflux about seventy times a day and one had more than twenty percent acid time, so always ask, abnormal for what age? The second is a positive symptom index used as a reason to operate: no index predicts outcome, and fundoplication fixes the numbers more reliably than the symptoms. The third is spasm on manometry in a small child: short latency is normal for size, spasm fell from fourteen percent to one percent with age adjustment, and an expert pediatric read is needed. The fourth is using FLIP to diagnose eosinophilic esophagitis: FLIP measures stiffness, biopsies diagnose the disease, and no decision should rest on FLIP alone.

The numbers worth carrying into clinic are these. Healthy preterm infants have about seventy-one reflux events a day, nearly three quarters weakly acidic and ninety percent reaching the upper esophagus. Non-acid reflux makes up forty-five to eighty-nine percent of pediatric episodes. Among two hundred and seventy abnormal pediatric studies, forty-five percent were abnormal by impedance only. There are no true pediatric normal values. The symptom index and the symptom association probability agree in about four of five studies,

and neither predicts surgical outcome. Under Chicago version four, outflow obstruction needs symptoms plus a supportive test, and IRP limits depend on the device. Adult criteria label two thirds of children abnormal, half after age adjustment. And on FLIP, a normal junction distensibility index is two or more, while in pediatric eosinophilic esophagitis a distensibility index below four and a half flags rings and impaction risk.

What to do differently in clinic.

Before ordering any esophageal function test, write down the question it must answer and the PPI status you need. Do not test the thriving infant with regurgitation, and do not let an infant's event count or acid time stand in for a diagnosis. When you read a report, look for the age-specific thresholds and the symptom indices, and weigh the indices only when symptoms were well marked. When a manometry report in a young child says spasm or outflow obstruction, ask for an expert pediatric review and a supportive test before anyone treats. And when a child with eosinophilic esophagitis keeps impacting despite good histology, measure distensibility, because the wall can stay stiff long after the eosinophils are gone.

End of chapter.